

Machine Learning Interview Handbook

Module 1 – First 10 Concept-Based Questions

Question 1: What is Machine Learning?

Answer

Machine Learning is a subset of Artificial Intelligence that enables computers to learn patterns from historical data and make predictions or decisions without being explicitly programmed for every situation. Instead of relying on fixed rules, the model learns from examples, identifies relationships in the data, and applies that knowledge to unseen data. Today it is widely used in recommendation systems, fraud detection, medical diagnosis, spam filtering and many other real-world applications.

Key Points

- Subset of AI
- Learns from historical data
- Generalizes to unseen data
- Predicts or makes decisions

Follow-up Questions

- How is ML different from Traditional Programming?
- Can ML work without data?

Question 2: Difference between Artificial Intelligence, Machine Learning and Deep Learning?

Answer

Artificial Intelligence (AI): AI is the broad field of creating systems capable of performing tasks that normally require human intelligence such as reasoning, planning and decision making.

Machine Learning (ML): ML is a subset of AI where systems learn patterns from historical data instead of relying only on manually written rules.

Deep Learning (DL): DL is a subset of ML that uses deep neural networks to automatically learn complex patterns from large datasets and is widely used in computer vision, speech recognition and LLMs.

Key Points

- $AI \supset ML \supset DL$
- DL uses neural networks
- DL generally requires more data and compute

Follow-up Questions •

Why is DL a subset of ML?

- Can AI exist without ML?

My Notes

Question 3: What are the Types of Machine Learning?**Answer**

Machine Learning is mainly divided into Supervised, Unsupervised, Semi-Supervised and Reinforcement Learning. The choice depends on the availability of labeled data and the business problem.

Key Points

- Know all four types
- Know one use case of each

Follow-up Questions

- Example of each type?

Question 4: Difference between Supervised and Unsupervised Learning?**Answer**

Supervised Learning uses labeled data to learn input-output relationships, whereas Unsupervised Learning works with unlabeled data to discover hidden patterns, groups or structures.

Key Points

- Labels vs No labels
- Prediction vs Pattern discovery

Follow-up Questions

- Algorithms of each?

Question 5: Difference between Classification and Regression?**Answer**

Classification predicts categorical outputs while Regression predicts continuous numerical values. The choice depends on the target variable.

Key Points

- Categorical vs Continuous

Follow-up Questions

- Why Logistic Regression is classification?

Question 6: What is a Dataset?

Answer

A dataset is a structured collection of observations used to train, validate and test machine learning models. Each row is generally a sample and each column is a feature.

Key Points

- Rows = Samples
- Columns = Features

Follow-up Questions

- Dataset vs Database?

My Notes

Question 7: What is a Feature and a Label?

Answer

Features are the input variables provided to the model, whereas the label is the target output that the model learns to predict.

Key Points

- Feature = X
- Label = y

Follow-up Questions

- Independent vs Dependent variable?

Question 8: Difference between Training, Validation and Test Dataset?

Answer

Training data is used to learn model parameters, validation data is used to tune hyperparameters, and test data is used only for final evaluation on unseen data.

Key Points

- Never tune on test set

Follow-up Questions

- Why not use test data for tuning?

My Notes

Question 9: Why do we split the Dataset?

Answer

Dataset splitting helps measure how well a model generalizes to unseen data. Without a proper split, the model may memorize the training data and produce misleading performance.

Key Points

- Detects overfitting
- Measures generalization

Follow-up Questions

- What if we don't split?

My Notes

Question 10: Difference between Training and Inference?

Answer

Training is the learning phase where model parameters are updated using training data. Inference is the prediction phase where the trained model makes predictions on new data without updating its parameters.

Key Points

- `fit()` vs `predict()`

Follow-up Questions

- Does inference update weights?

My Notes

Machine Learning Interview Handbook

Module 1 - Questions 31 to 40

31. What is Structured Data?

Answer

Structured data is data organized in rows and columns with a fixed schema. It is easy to store, search and analyze using databases.

Example

Customer table with Name, Age and Salary stored in SQL.

Key Points

- Fixed format

- Rows & Columns
- Easy to query

Follow-up Questions

What are examples of structured data?

Which databases store it?

My Notes

32. What is Unstructured Data?

Answer

Unstructured data does not follow a predefined format. It includes text, images, videos, audio and documents.

Example

Photos uploaded on Instagram or medical X-ray images.

Key Points

- No fixed schema
- Difficult to analyze
- Most real-world data

Follow-up Questions Give examples.

Why is it difficult to process?

My Notes

33. What is Semi-Structured Data?

Answer

Semi-structured data does not follow a strict tabular format but contains tags or key-value pairs that provide organization.

Example

JSON, XML and HTML files.

Key Points

- Flexible structure

- Metadata available
- Easier than unstructured

Follow-up Questions Is

JSON structured?

Where is XML used?

My Notes

34. Difference between Structured, Semi-Structured and Unstructured Data?

Answer

Structured data has a fixed schema, semi-structured data has a flexible schema with tags, and unstructured data has no predefined organization.

Example

SQL table vs JSON vs Image.

Key Points

- SQL → Structured
- JSON/XML → Semi
- Images/Text → Unstructured

Follow-up Questions

Which type is most common today?

My Notes

35. What is Batch Learning?

Answer

Batch Learning trains the model using the entire dataset at once. To learn new data, the model is retrained from scratch.

Example

Monthly sales prediction model retrained every month.

Key Points

- Entire dataset
- Offline training
- Periodic retraining

Follow-up Questions

When is batch learning preferred?

My Notes

36. What is Online Learning?

Answer

Online Learning updates the model continuously as new data arrives instead of retraining from scratch.

Example

Fraud detection model learning from new transactions daily.

Key Points

- Continuous updates
- Good for streaming data
- Fast adaptation

Follow-up Questions

Where is online learning used?

My Notes

37. Difference between Batch Learning and Online Learning?

Answer

Batch learning learns from all available data together, whereas online learning updates incrementally whenever new data arrives.

Example

E-commerce recommendation systems often use online learning.

Key Points

- Batch = Periodic
- Online = Continuous
- Streaming support

Follow-up Questions

Which is better for real-time systems?

My Notes

38. What is Data Leakage?

Answer

Data leakage occurs when information from the test data or future data accidentally becomes available during training, causing unrealistically high performance.

Example

Applying scaling before train-test split is a common leakage example.

Key Points

- Unrealistic accuracy
- Wrong evaluation
- Must be avoided

Follow-up Questions

How can leakage be prevented?

My Notes

39. What are the Applications of Machine Learning?

Answer

Machine Learning is used in healthcare, finance, e-commerce, cybersecurity, agriculture, transportation and natural language processing to automate predictions and decision-making.

Example

Spam detection, Netflix recommendations and disease prediction.

Key Points

- Healthcare
- Finance
- Recommendation Systems

Follow-up Questions

Give five real-world applications.

My Notes

40. What are the Challenges or Limitations of Machine Learning?

Answer

Machine Learning requires high-quality data, sufficient computational resources and careful evaluation.

Poor data, overfitting, bias, interpretability and privacy issues are common challenges.

Example

Biased hiring data can produce unfair recruitment models.

Key Points

- Data quality
- Overfitting
- Interpretability
- Bias & Privacy

Follow-up Questions

What is the biggest challenge in ML projects?

My Notes

Machine Learning Interview Handbook

Module 1 – Questions 21–30

21. What is Bias in Machine Learning?

Answer

Bias is the error caused when a model makes overly simple assumptions about the data. High bias models fail to learn important patterns and usually underfit.

Example: Using Linear Regression to model a highly non-linear relationship may result in high bias.

Key Points

- Simplified assumptions
- Causes underfitting
- High training error

Follow-up Questions What is

High Bias?

How can we reduce bias?

My Notes

22. What is Variance in Machine Learning?

Answer

Variance measures how much a model's predictions change when trained on different datasets. High variance models memorize training data and overfit.

Example: A very deep Decision Tree often memorizes the training data and performs poorly on unseen data.

Key Points

- Sensitive to training data
- Causes overfitting
- Low training error

Follow-up Questions

How can we reduce variance?

Give an example of a high-variance algorithm.

My Notes

23. What is the Bias-Variance Tradeoff?

Answer

The Bias-Variance Tradeoff is the balance between underfitting and overfitting. A good model achieves low prediction error by maintaining the right balance.

Example: Increasing Decision Tree depth decreases bias but increases variance.

Key Points

- Balance between bias and variance
- Avoid underfitting
- Avoid overfitting

Follow-up Questions

Why can't both be zero?

Which models usually have high bias?

My Notes

24. What is Noise in Machine Learning?

Answer

Noise is unwanted or incorrect information in the dataset that does not represent real patterns.

Example: Wrong labels, typing mistakes, sensor errors, or missing values.

Key Points

- Random errors
- Incorrect information
- Can cause overfitting

Follow-up Questions

Can noise be removed completely?

Give examples.

My Notes

25. What is Model Complexity?

Answer

Model complexity refers to a model's ability to learn complicated patterns. Complex models may overfit, while simple models may underfit.

Example: A Random Forest is generally more complex than Linear Regression.

Key Points

- Simple vs Complex
- Related to overfitting
- Depends on algorithm

Follow-up Questions

Decision Tree vs Linear Regression?

Why is complexity important?

My Notes

26. What is Prediction?

Answer

Prediction is the process of using a trained model to estimate outputs for new unseen data.

Example: Predicting whether a patient has heart disease using a trained model.

Key Points

- Uses trained model
- Unseen data
- No learning occurs

Follow-up Questions

Is prediction part of training?

Which function is commonly used?

My Notes

27. What is a Target Variable?

Answer

The target variable is the output the model learns to predict. It is also called the label or dependent variable.

Example: In house price prediction, 'Price' is the target variable.

Key Points

- Output variable

- Label (y)
- Used in supervised learning

Follow-up Questions Feature vs
Target?

Is there a target in Unsupervised Learning?

My Notes

28. What is an Independent Variable?

Answer

An independent variable (feature) is an input used to predict the target variable.

Example: Age, Salary and Experience are features for predicting loan approval.

Key Points

- Input variable
- Feature (X)
- Used for prediction

Follow-up Questions

Independent vs Dependent variable?

Give an example.

My Notes

29. What is a Machine Learning Pipeline?

Answer

A Machine Learning Pipeline automates the sequence of preprocessing, training, evaluation and prediction.

Example: StandardScaler → PCA → Random Forest in a single sklearn Pipeline.

Key Points

- End-to-end workflow
- Automates ML process
- Reduces data leakage

Follow-up Questions Why use
Pipelines?

Which library provides them?

My Notes

30. What are the Steps in a Typical Machine Learning Project?

Answer

A typical ML workflow includes Problem Understanding, Data Collection, Data Cleaning, EDA, Feature Engineering, Data Splitting, Model Selection, Training, Evaluation, Hyperparameter Tuning, Deployment, and Monitoring.

Example: Heart Disease Prediction: Collect patient data → Clean → Train Random Forest → Evaluate → Deploy with Flask.

Key Points

- End-to-end ML lifecycle
- Data → Model → Deployment
- Continuous monitoring

Follow-up Questions

Which step takes the most time?

Why is monitoring needed?

My Notes

Machine Learning Interview Handbook

Module 1 – Questions 11–20

11. What is an Algorithm?

Answer

An algorithm is a step-by-step procedure or set of rules used to solve a particular problem. In Machine Learning, an algorithm learns patterns from data and creates a model that can make predictions on unseen data. Different algorithms are suitable for different problems depending on the data and requirements.

Key Points

- Step-by-step procedure
- Learns from data
- Creates a predictive model

Follow-up Questions

Is an algorithm the same as a model?

Name some ML algorithms?

My Notes

12. Difference between Algorithm and Model?

Answer

An algorithm is the learning procedure that analyzes data and learns patterns, while a model is the final output produced after training. For example, Decision Tree is an algorithm, whereas the trained Decision Tree after calling fit() is the model used for prediction.

Key Points

- Algorithm learns
- Model predicts
- fit() creates the model

Follow-up Questions

Which one is deployed in production?

Can one algorithm produce multiple models?

My Notes

13. What is Training Data?

Answer

Training data is the portion of the dataset used to teach the machine learning model. During training, the algorithm identifies relationships between input features and target labels by adjusting its internal parameters.

Key Points

- Used for learning
- Largest split
- Updates parameters

Follow-up Questions

What if the training dataset is too small?

Can training data contain noise?

My Notes

14. What is Testing Data?

Answer

Testing data is a separate portion of the dataset used only after training to evaluate the model's performance on unseen data. It provides an unbiased estimate of real-world performance.

Key Points

- Unseen data
- Final evaluation
- Never used during training

Follow-up Questions

Why should the test set remain untouched?

What if we use the test set for tuning?

My Notes

15. What are Parameters in Machine Learning?

Answer

Parameters are internal values learned automatically during training. They determine how the model makes predictions.

Key Points

- Learned automatically
- Updated during training
- Affect predictions

Follow-up Questions Give examples

of parameters.

Who learns the parameters?

My Notes

16. What are Hyperparameters?

Answer

Hyperparameters are configuration settings defined before training begins. They are not learned from data and control the learning process.

Key Points

- Set before training
- Not learned automatically
- Control learning

Follow-up Questions

Difference between parameters and hyperparameters?

Give three examples.

My Notes

17. Difference between Parameters and Hyperparameters?

Answer

Parameters are learned automatically from training data, whereas hyperparameters are chosen before training starts and control how learning happens.

Key Points

- Parameters: learned
- Hyperparameters: predefined

Follow-up Questions

Which one does GridSearchCV optimize?

Give one example of each.

My Notes

18. What is Generalization?

Answer

Generalization is the ability of a model to perform well on new unseen data instead of memorizing the training data.

Key Points

- Performs well on unseen data
- Goal of ML
- Opposite of memorization

Follow-up Questions

Why is generalization important?

Which problem affects generalization?

My Notes

19. What is Overfitting?

Answer

Overfitting occurs when a model learns both useful patterns and noise in the training data, resulting in poor performance on unseen data.

Key Points

- Memorizes training data
- High training accuracy
- Poor testing accuracy

Follow-up Questions

How can we reduce overfitting?

Which algorithms are prone to overfitting?

My Notes

20. What is Underfitting?

Answer

Underfitting occurs when a model is too simple to learn the underlying patterns, leading to poor performance on both training and testing data.

Key Points

- Model too simple
- Poor training accuracy
- Poor testing accuracy

Follow-up Questions

How can underfitting be reduced?

Give one real-world example?

My Notes
